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Clinical utility of correction factors for febrile young infants with traumatic lumbar punctures

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Abstract

Objectives: Correction factors have been proposed for traumatic lumbar punctures (LPs) in febrile young infants. However, no studies have assessed their diagnostic utility. We sought to determine the proportion of traumatic LPs safely reclassified as low risk for bacterial meningitis using recently derived white blood cell (WBC) and protein correction factors.

Methods: We retrospectively analyzed traumatic LPs among all febrile infants ≤60 days old at two tertiary paediatric hospitals from 2006 through 2018. Traumatic LPs were defined as ≥10,000 RBCs/mm^3. Abnormal cerebrospinal fluid (CSF) WBCs and protein were adjusted downward using a newly derived correction factor (877 red blood cells [RBCs]: 1 WBC), three commonly used correction factors (500 WBCs: 1 RBC; 1,000 WBCs: 1 WBC; peripheral RBCs: WBCs), and a newly derived protein correction factor (1,000 RBCs: 0.011 g/L protein).

Results: There were 437 traumatic LPs including 357 (82%) with pleocytosis and 4 (0.9%) with bacterial meningitis. Overall, fewer infants were classified as having CSF pleocytosis using 877:1 and 1,000:1 ratios (38% and 43%, respectively), with 100% sensitivity and negative predictive value, and improved specificity (63% for 877:1, 58% for 1,000:1 ratios versus 19% for uncorrected counts). Among infants with pleocytosis, 877:1 and 1,000:1 ratios reclassified 191 (54%) and 171 (48%) as normal with no misclassified bacterial meningitis cases. Ratios of 500:1 and peripheral RBC:WBC misclassified 1 infant that had bacterial meningitis. Corrected CSF protein outperformed uncorrected protein in specificity but did not add diagnostic value following WBC-based correction.

Conclusions: Correction ratios of 877:1 and 1,000:1 safely reclassified half of all febrile infants ≤60 days. These corrections should be considered when interpreting CSF results of traumatic LPs.

Keywords: *Fever; Meningitis; Serious bacterial infection; Spinal tap*

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Nearly 2% of all term newborns will be evaluated for fever in an emergency department (ED) within their first months of life (1). Published guidelines recommend that infants ≤60 days old undergo evaluation for sepsis and risk stratified using a combination of clinical factors and laboratory values frequently including lumbar puncture (LP) for cerebrospinal fluid (CSF) testing (2). CSF cultures require up to 48 hours to reliably exclude bacterial meningitis (3–5), and consequently, management decisions regarding empiric antibiotic treatment and hospitalization must be made on the basis of the initial CSF parameters. When a LP is traumatic, blood may be iatrogenically introduced into the CSF, often rendering results uninterpretable by elevating both the WBC count and the protein content. Traumatic LPs for febrile young infants are reported to occur at a rate of 10% to 65% depending on provider experience (6,7), and infants with traumatic LPs are more likely to receive intravenous antibiotics and be hospitalized (8). A reliable CSF correction strategy could reduce hospitalization and antibiotic exposure for febrile young infants with traumatic LPs. Historically, several ratio-based correction methods have been used to identify low-risk infants with traumatic LPs yet the use of ratio-based CSF correction factors for traumatic LPs has remained controversial for decades.

Recently, 2 large studies from 20 US paediatric EDs derived novel correction factors to facilitate the interpretation of CSF results in febrile infants ≤60 days old with traumatic LPs (9,10). Briefly, these newly derived correction factors suggest to deduct 1 WBC for every 877 RBCs, and to deduct 0.11 g/dL protein for every 1,000 RBCs/mm^3. When applied, these correction factors improved specificity for discriminating infants with bacterial meningitis compared to uncorrected CSF results. However, studies assessing their diagnostic utility in clinical practice are entirely lacking. Moreover, the importance of external validation for derived models is well established and essential before widespread use (11,12).

We sought to evaluate the clinical utility of these novel ratio-based WBC (9) and protein (10) correction factors, when applied in an external population of febrile young infants with traumatic LPs. The primary objective was to determine the proportion of infants with traumatic LPs that could be accurately reclassified as low risk for bacterial meningitis using both recently derived correction factors. Secondary objectives were to (1) determine the number of infants that would be misclassified by these correction factors and (2) to compare the test characteristics of these newly described correction factors to older, commonly used ratio-based correction methods (13).

METHODS

Study design and setting

We undertook a multicentre retrospective cohort study at two urban tertiary paediatric hospitals in Montreal, Canada, each with a census of over 80,000 annual ED visits. The study protocol was approved by both institutional Research Ethics Boards.

Selection of participants

All infants were eligible for inclusion if they were aged ≤60 days and presented to either ED between January 1, 2006 and December 31, 2018 and underwent an LP. Infants were included if they had both CSF culture and cell counts obtained within 24 hours of ED presentation. Infants missing either a CSF culture or cell count were excluded. Additionally, all infants that received antibiotics before LP, infants that had any previous LP attempts during the same febrile episode, and infants with a ventriculoperitoneal shunt or other implanted intracranial devices were excluded. Infants with greater than one eligible ED visit were included for each encounter.

Eligible subjects were identified using the microbiological database of both institutions to capture all infants for whom a CSF culture was successfully obtained. To determine the denominator of infants with traumatic LPs meeting inclusion, the electronic medical record was reviewed, and additional data were collected including demographic information (age and sex), triage temperature, diagnosis, disposition, antibiotics received and timing with relation to LP, laboratory data (urinalysis results, CBC counts, CSF cell counts, protein and glucose), and complete microbiological data (CSF, blood, and urine culture results, herpes simplex virus [HSV], and enterovirus testing, when obtained).

Study definitions

As in the derivation studies, traumatic LP was defined as a CSF specimen with ≥10,000 RBC/mm^3 and without an upper limit, reflecting a level of blood infiltration rending CSF results difficult to interpret (9,10). Thresholds for normal CSF WBCs and protein were age specific as described previously (9,10). CSF pleocytosis was defined as a CSF WBC count ≥20 cells/mm^3 for infants aged 0 to 28 days and ≥10 cells/mm^3 for infants aged 29 to 60 days (9), and a normal CSF protein value was defined as ≤1.15 g/L for infants aged ≤28 days and ≤0.89 g/L for infants aged 29 to 60 days (10).

Culture-proven meningitis was defined as a bacterial pathogen in CSF culture or positive bacterial Gram stain. The following microorganisms were classified as CSF contaminants a priori: coagulase-negative staphylocci, *Micrococcus* spp., *Propionibacterium spp.*, and *Corynebacterium spp.*, or any other bacterial pathogen treated clinically as a contaminant upon chart review by a microbiologist. HSV and enterovirus CSF infections were defined as a positive CSF HSV and enterovirus polymerase chain reaction (PCR) test, respectively (9,10).

Data analysis

Standard descriptive statistics were used to describe clinical characteristics of included infants with traumatic LPs.

Uncorrected CSF WBCs and protein were adjusted downward using RBC ratio-based correction methods. We evaluated the performance of the recently derived CSF WBC (877:1) (9) and protein (1,000 RBCs: 0.011 g/L protein) (10) correction factors, as well as commonly used WBC ratios (500:1, 1,000:1, and peripheral RBC:WBC) (13). CSF WBCs were corrected by subtracting 1 WBC for every 500, 877, or 1,000 CSF RBCs, or by deducting the number of WBCs determined by the ratio of RBC-to-WBC in the peripheral blood. Similarly, CSF protein was corrected by subtracting 0.011 g/L from the observed CSF protein for every 1,000 CSF RBCs. A composite correction was also evaluated by reclassifying CSF meeting normal parameters following correction by both 877:1 WBC and 0.011 g/L protein. Test characteristics were calculated for unadjusted and adjusted CSF values for predicting culture-proven bacterial meningitis (area under the receiver-operator curve [AUC], sensitivity, specificity, and negative predictive value [NPV]). Additionally, we conducted sensitivity analyses to evaluate correction factors with normal CSF WBC cut-points of >18 and >15 cells/mm^3 for infants aged 0 to 28 days, and >9 and >4 cells/mm^3 for infants aged 29 to 60 days to reflect other common definitions (14,15). Statistical analyses were performed using Stata v.14.1 (StataCorp, College Station, TX).

RESULTS

Characteristics of study subjects

Overall, 5,172 infants ≤60 days old underwent LP with CSF cultures available during the 13-year study period (Supplementary Figure 1). Among these, 4,912 (95.0%) had both CSF culture and cell counts available for analysis, and our study population included 437 (8.9%) infants with traumatic LPs. Demographic characteristics of included infants are displayed in Table 1 and did not differ between the two centres. Most infants with traumatic LPs were admitted to hospital (96.2%) and 64.3% of infants were ≤28 days old. Among included infants, 249 (57.0%) had a triage temperature ≥38.0°C, and an additional 169 (38.7) had admitting diagnoses of febrile illnesses or SBIs. CSF testing for HSV and enterovirus was infrequent (48 and 59 infants, respectively). Bacterial meningitis occurred in four (0.92%) infants, and the pathogens identified were Group B *streptococcus* (three cases) and *Escherichia coli* (one case). No pathogens required adjudication by microbiologists outside those defined as contaminants a priori.

Main results

All infants with traumatic LPs, when using WBC cut-offs of ≥20 cells/mm^3 for infants aged 0 to 28 days and ≥10 cells/mm^3 for infants aged 29 to 60 days (9), 357/437 (82%; 95%CI: 78% to 85%) had an uncorrected CSF WBC pleocytosis (Table 2). The 877:1 correction factor reclassified 191/357 infants (54%; 95%CI: 48% to 59%) as not having CSF pleocytosis without misclassifying any that had bacterial meningitis. The 1,000:1 correction factor performed similarly, reclassifying 171/357 infants (48%; 95%CI: 43% to 53%), with none misclassified. Overall, both performed with 100% (95%CI: 40% to 100%) sensitivity and NPV. The 500:1 and the peripheral RBC:WBC ratios reclassified

Table 1. Demographic information for study population of infants aged 0 to 60 days old with traumatic LPs (≥10,000 RBC/mm^3); N=437

Characteristics		
Age in days	All*	23 (14–34)
	0–28	282/437 (64.5)
	29–60	155/437 (35.5)
Sex, n/N (%)	Male infant	240/437 (54.9)
Clinical	Triage temperature, °C*	38.1 (37.4–38.6)
	CBC WBC count, cells/mm^3*	10.9 (7.9–15.4)
	Admitted, n/N (%)	420/437 (96.1)
	Length of stay for admitted, days*	2.6 (2.1–3.8)
Microbiology results by age category†		
Bacterial meningitis, n/N (%), days	0–28	2/282 (0.7)
	29–60	2/155 (1.3)
HSV meningoencephalitis, n/N tested (%), days	0–28	0/40 (0.0)
	29–60	0/8 (0.0)
Enteroviral meningitis, n/N tested (%), days	0–28	15/44 (34.1)
	29–60	6/15 (40.0)

HSV Herpes simplex virus; LP Lumbar puncture; RBC Red blood cell; WBC White blood cell.

*Median (interquartile range).

†Numbers do not sum because microbiologic testing was not available for all patients.

Table 2. Traumatic LPs with CSF pleocytosis (>20 cells/mm^3 for infants ≤28 days old; >10 cells/mm^3 for infants aged 29 to 60 days old); uncorrected and using different CSF WBC correction strategies (N=437 infants, four bacterial meningitis)

Strategy	CSF Pleocytosis N (%)	N (%) Reclassified normal after correction	N (%) Misclassified cases of bacterial meningitis	Sensitivity % (95% CI)	Specificity, % (95% CI)	NPV % (95% CI)	AUC
Uncorrected CSF WBC count	357 (81.7)	Reference	0 (0.0)	100.0 (39.8–100.0)	18.5 (14.9–22.1)	100.0 (95.5–100.0)	0.591 (0.321–0.861)
Derived 877:1 correction	166 (38.0)	191 (53.5)	0 (0.0)	100.0 (39.8–100.0)	62.6 (57.8–67.1)	100.0 (98.6–100.0)	0.804 (0.672–0.936)
1,000:1 correction	186 (42.6)	171 (47.9)	0 (0.0)	100.0 (39.8–100.0)	58.0 (53.1–62.7)	100.0 (98.5–100.0)	0.786 (0.642–0.930)
500:1 correction	96 (22.0)	261 (73.1)	1 (0.4)	75.0 (19.4-99.4)	78.5 (74.4–82.3)	99.7 (98.4–99.9)	0.855 (0.804–0.965)
Peripheral ratio RBC:WBC	64 (14.6)	293 (82.1)	1 (0.3)	75.0 (19.4-99.4)	85.9 (82.2–89.0)	99.7 (98.5–99.9)	0.932 (0.868–0.978)

AUC Area under the receiver-operator curve; CI Confidence interval; CSF Cerebrospinal fluid; NPV Negative predictive value; RBC Red blood cell; WBC White blood cell.

Table 3. Traumatic LPs with abnormal CSF protein (>1.15 for infants ≤28 days old; >0.89 g/L for infants aged 29 to 60 days old); uncorrected and using derived CSF protein correction strategy (N=95 infants, three bacterial meningitis)

Strategy	Abnormal CSF protein N (%)	N (%) Reclassified normal after correction	N (%) Misclassified cases of bacterial meningitis	Sensitivity % (95% CI)	Specificity, % (95% CI)	NPV % (95% CI)	AUC
Uncorrected CSF protein	262 (66.3)	Reference	0 (0.0)	100.0 (29.2–100)	33.9 (29.3–38.9)	100.0 (97.3–100.0)	0.671 (0.452–0.889)
CSF protein correction	133 (33.7)	129 (49.2)	0 (0.0)	100.0 (29.2–100)	66.8 (61.9–71.5)	100.0 (98.6–100.0)	0.835 (0.722–0.947)

AUC Area under the receiver-operator curve; CI Confidence interval; CSF Cerebrospinal fluid; LP Lumbar puncture; NPV Negative predictive value.

261/357 (73%; 95%CI: 68% to 78%) and 293/357 (82%; 95%CI: 78% to 86%) of infants, respectively. Both of these correction ratios misclassified the same 12-day-old infant with Group B streptococcus meningitis and bacteremia (Supplementary Table 1). Consequently, sensitivity for both these correction ratios was 75% (95%CI: 19% to 99%). All WBC correction ratios demonstrated improved specificity and AUC compared to uncorrected values. We performed sensitivity analyses using commonly used normal CSF cut-points; >18 and >15 cells/mm^3 for infants aged 0 to 28 days, and >9 and >4 cells/mm^3 for infants aged 29 to 60 days. All WBC correction factors yielded similar results at these other thresholds (Supplementary Tables 2 and 3). Using the most conservative of these age-specific cut points, the 877:1 correction factor reclassified nearly 50% (190/384) of all infants with pleocytosis as normal without any misclassifications.

In terms of CSF protein, 395 infants with traumatic LPs had protein results available for analysis (Table 3). Among these, 262 (66%; 95%CI: 61% to 71%) had abnormal uncorrected CSF protein values. The newly derived protein correction factor reclassified 129/262 (49%; 95%CI: 43% to 55%) infants as not having elevated protein while misclassifying none. All test characteristics for corrected protein were as good or better than uncorrected protein values alone, however, protein correction did not add diagnostic utility to the 877:1 WBC correction factor when used in composite (sensitivity 100%, 95%CI: 40% to 100%; specificity 46%, 95%CI: 41% to 51%; NPV 100%, 95%CI: 98.2% to 100.0%; AUC 0.71, 95%CI: 0.52 to 0.90).

DISCUSSION

We assessed recently derived WBC and protein correction factors for the interpretation of traumatic LPs in an external population of infants ≤60 days old. WBC correction ratios of 877:1 and 1,000:1 safely reclassified about half of all febrile infants with traumatic LPs as not having CSF pleocytosis, while 500:1 and peripheral RBC:WBC ratios misclassified one of four infants with bacterial meningitis. CSF protein correction offered enhanced specificity compared to uncorrected protein values. However, protein correction did not improve diagnostic utility beyond WBC correction alone. These findings add to the literature using WBC correction factors to interpret traumatic LPs for febrile young infants (16).

To our knowledge, this is the first external validation of the newly derived WBC and protein correction factors, and the first study to evaluate their use in combination. Infants in our study were demographically similar to the derivation cohort, including similar rates of bacterial meningitis (0.92%) (9,10). In this study, there were only four infants with traumatic LPs and culture-confirmed bacterial meningitis. Therefore, despite finding test performances similar to those reported in the derivation studies, we observe wide CIs. Importantly though, given that bacterial meningitis is a rare outcome, the primary objective of this study was to assess the clinical utility of WBC and protein correction factors, alone and in combination, to determine the proportion of infants with traumatic LPs that could be safely reclassified as low risk for bacterial meningitis. As reported in the derivation study, the 877:1 correction ratio would have reclassified 191/357 infants (54%) with no misclassifications at two centres over 13 years. Interestingly, the 500:1 and peripheral RBC:WBC correction ratios performed with the highest AUC, owing to highest specificity. However, given the consequences of incorrectly reclassifying an infant with bacterial meningitis which occurred once with these ratios, a correction with high sensitivity and NPV should be prioritized.

The use of uncorrected CSF WBC counts inherently leads to empiric antibiotic and/or antiviral exposure, as well as hospitalization for many infants without bacterial meningitis (8). Unnecessary broad-spectrum antibiotic use exposes patients to potential harms and drives pathogen resistance (17). Prolonged hospital admission presents risks of nosocomial infections and medical errors, as well as costs borne by healthcare systems and families. Utilizing the corrected CSF WBC should help inform shared-decision making to balance risks and benefits and could obviate the need for hospitalization and parenteral antibiotics in otherwise low-risk infants. In our analysis, the only misclassified infant was <28 days old, and most clinical decision tools recommend empiric antibiotics and hospitalization regardless of LP results in this age group (18,19). For infants >28 days old otherwise meeting low-risk criteria, CSF correction most stands to reduce rates of hospitalization and antibiotic use (19).

Application of any correction factor will reduce the number of infants classified as having CSF pleocytosis and thus increase specificity at the cost of decreased sensitivity for bacterial meningitis. In the derivation study by Lyons et al. the 877:1 WBC correction factor reclassified 1,305/2,880 (45%) infants with traumatic LPs (defined as >10,000 cells/mm^3 CSF RBCs), at the cost of misclassifying 7 of 33 infants with bacterial meningitis. Importantly, only 1 misclassified infant was >28 days old. In a large multicentre study conducted across 150 neonatal intensive care units of 2,519 infants ≤30 days old with traumatic LPs (defined as ≥500 cells/mm^3 CSF RBCs), adjustment of CSF WBCs did not aid in the diagnosis of culture-proven meningitis (13). However, this study used a markedly different definition of traumatic LP and did not include infants >30 days old. The CSF protein correction factor also derived by Lyons et al.(10) is in keeping with a previously proposed protein correction factor (20). This study suggests that protein correction adds no significant diagnostic value when used alone or in combination with WBC correction, compared to 877:1 WBC correction alone.

For validation purposes, we used the same thresholds for CSF pleocytosis as Lyons et al. (0 to 28 days old, ≥20 WBC/mm^3; 29 to 60 days old, ≥10 WBC/mm^3); however, other clinical thresholds have also been proposed. Sensitivity analyses using other common thresholds for CSF pleocytosis (14,15) demonstrated that all WBC correction factors performed similarly at each CSF threshold. Of note, the 877:1 WBC correction reclassified half of all infants with pleocytosis without any misclassifications, regardless of CSF WBC cut-point used.

Our study has limitations. First, we could not include infants for whom there was insufficient CSF for culture and cell counts. It is possible that some infants with bacterial meningitis had grossly bloody CSF samples that were not sent for cell count. However, only 260/5,172 (5%) of all eligible LPs did not have accompanying cell counts. As in the derivation studies, we did not set an upper limit for CSF RBC counts (9,10), thus we cannot determine if some samples consisted only of peripheral blood, although all samples were identified by the clinician as being CSF and peripheral blood alone and would be expected to clot. We did not evaluate CSF pleocytosis in the prediction of viral meningitis as testing was rare. Owing to the retrospective design, we cannot know if all infants presented with fever. However a majority (418/437; 96%) had fever recorded at triage or an admitting diagnosis consistent with a febrile illness or SBI, and in all cases, the treating physician undertook testing to rule out bacterial meningitis. Confirmed bacterial meningitis cases were few; however, it represents all cases among infants with traumatic LPs at two large, high-volume paediatric hospitals over a 13-year study period, highlighting the utility of a correction that could reduce uninterpretable and false positives LPs. Finally, we included only infants ≤60 days old and cannot extrapolate findings to older infants and children with traumatic LPs.

CONCLUSION

Traumatic LPs are common and render interpretation of CSF cell counts in febrile infants difficult. As a result, many infants are hospitalized and receive intravenous antibiotics unnecessarily. CSF correction by either 877:1 or 1,000:1 would have safely reclassified approximately half of all infants with traumatic LPs as low risk; 877:1 performed with the best overall test characteristics, while 1,000:1 may be more intuitive to apply. Findings support the use CSF WBC correction for traumatic LPs to more effectively risk stratify febrile young infants, particularly those aged 29 to 60 days, to potentially reduce hospitalizations and broad-spectrum antibiotic exposure.

SUPPLEMENTARY DATA

Supplementary data are available at *Paediatrics & Child Health* Online.

Supplementary Figure 1. Flow diagram of included patients

Supplementary Table 1. Characteristics of the infant with bacterial meningitis misclassified by 500:1 and peripheral RBC:WBC ratio correction strategies.

Supplementary Table 2. Traumatic LPs with CSF pleocytosis (>18 cells/mm^3 for infants ≤28 days old; >9 cells/mm^3 for infants 29-60 days old); uncorrected and using different CSF WBC correction strategies. N = 437 infants, 4 bacterial meningitis.

Supplementary Table 3. Traumatic LPs with CSF pleocytosis (>15 cells/mm^3 for infants ≤28 days old; >4 cells/mm^3 for infants 29-60 days old); uncorrected and using different CSF WBC correction strategies. N = 437 infants, 4 bacterial meningitis.

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